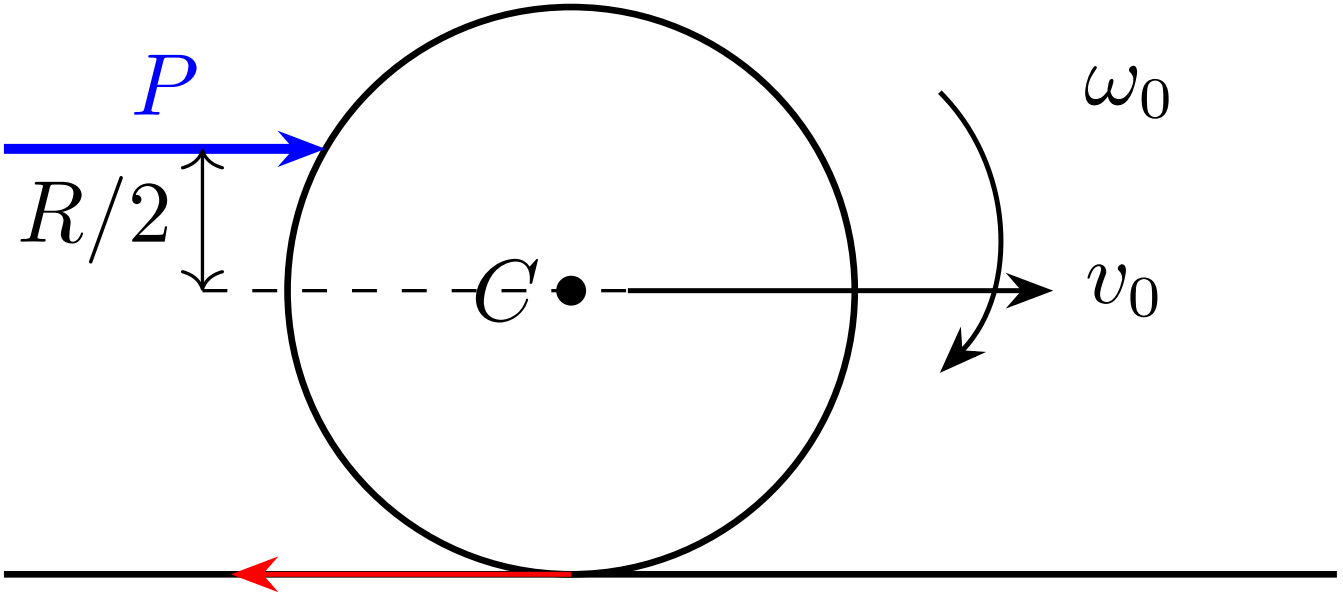

$$u_0 = -P/(4m)$$